

## CHEMICAL PHYSICS

Ultrafast reversible photoconductivity in 2D MoTe<sub>2</sub>/Pt van der Waals heterostructureYe Tao<sup>1</sup>, Chengyun Hong<sup>1</sup>, Ji-Hee Kim<sup>2\*</sup>

Two-dimensional (2D) materials, particularly transition metal dichalcogenides, have exceptional optoelectronic properties, making them highly promising for next-generation photonic integrated circuits. Despite great advancements in 2D optoelectronic devices, achieving ultrafast and controllable photoconductivity polarity inversion with a single device remains a fundamental challenge due to the static nature of built-in electric fields at metal/2D material interfaces. This study demonstrates a transient electric field reversal at the MoTe<sub>2</sub>/Pt Schottky junction, enabling photoconductivity inversion from negative to positive within 100 ps. By applying ultrafast photocurrent detection, a minimal voltage variation (10 mV) precisely controls this transition, and a device with a remarkable photocurrent response time of 3.8 ps is proposed. This work advances the design of ultrafast, tunable photodetectors, offering potential applications in high-speed optical communication, ultrafast imaging, and quantum information processing.

## INTRODUCTION

In recent years, two-dimensional (2D) materials, particularly transition metal dichalcogenides, have garnered extensive attention as promising candidates for next-generation optoelectronic devices. Their extraordinary electronic and optical properties, including high detectivity (1, 2), broadband spectral response (3, 4), and ultrafast photoresponse (5–7), enable groundbreaking advancements in photodetector technologies, even at nanometer-scale dimensions. These remarkable attributes make 2D materials well suited for integration into photonic integrated circuits (PICs), fulfilling the growing demand for ultrafast, highly tunable, and energy-efficient optoelectronic components (8). A critical factor in realizing high-performance 2D material-based photodetectors is the nature of the metal-semiconductor contact and the associated electric field at the interface. Conventionally, this electric field is considered time-invariant (steady state), and most studies have focused on device performance under such conditions. However, achieving ultrafast response times, precise electrical tunability, and novel photoconductivity phenomena, such as polarity inversion under light illumination, remains a formidable challenge within the constraints of a static electric field. Under steady-state conditions, the photoconductivity remains fixed, making it challenging to achieve negative photoconductivity inversion (Fig. 1A). Consequently, introducing new mechanisms or external components becomes essential to enable this capability, such as tuning the incident light wavelength (9), gate control (including the ferroelectric dielectrics-induced polarization field) (10–13), or adjusting environmental parameters like gas pressure (14). While effective, these methods often have limitations, including high energy consumption, slow operational speeds, complex fabrication requirements, and additional circuit components for edge detection and wavelength switching. These constraints present substantial challenges for practical applications, necessitating alternative strategies to achieve ultrafast and energy-efficient photoconductivity control.

Unlike steady-state conditions, ultrafast light excitation in 2D materials can induce rapid electric field inversion due to carrier accumulation on the picosecond or femtosecond scale (15, 16). This effect is further enhanced by flexible band alignment in van der Waals heterostructures (17, 18) and light-induced band renormalization (19, 20), enabling rapid carrier movement reversal within an ultrashort time frame (Fig. 1B). For instance, Zeng *et al.* demonstrated picosecond-scale response times and dynamic polarity switching of photocurrents in MoS<sub>2</sub>/WSe<sub>2</sub> n-n junction heterostructures under varying bias conditions (21). Although earlier polarity inversion studies relied on interlayer charge transfer or external gating, our results demonstrate that the metal/2D semiconductor interfaces, widely used in practical devices and characterized by strong built-in fields, can also support electric field-driven polarity modulation in an ultrashort timescale. Given the fundamental importance of such interfaces in real-world optoelectronic devices, the transient electric field effects uncovered here offer a pathway toward reconfigurable, high-speed photodetectors.

In this work, we use ultrafast photocurrent (UFPC) detection on a MoTe<sub>2</sub>/Pt Schottky junction to demonstrate a transient electric field-driven polarity inversion in photoconductivity. We observe a rapid transition from negative to positive photoconductivity within a temporal window of just 100 ps, governed by the dynamic interplay between photoexcited carriers and the internal electric field at the 2D material/metal interface. Notably, the photoconductivity transition can be finely controlled by minimal adjustments to the applied reverse bias voltage, achieving an exceptional sensitivity of 10 mV. Furthermore, we propose an innovative device capable of responding to alternating voltage inputs, achieving an extraordinary photocurrent response time of 3.8 ps, placing it among the fastest reported optoelectronic systems. This combination of ultrafast response and high tunability represents a paradigm shift in the design and functionality of next-generation photodetectors.

## RESULTS

## Light-induced photoconductivity inversion detection

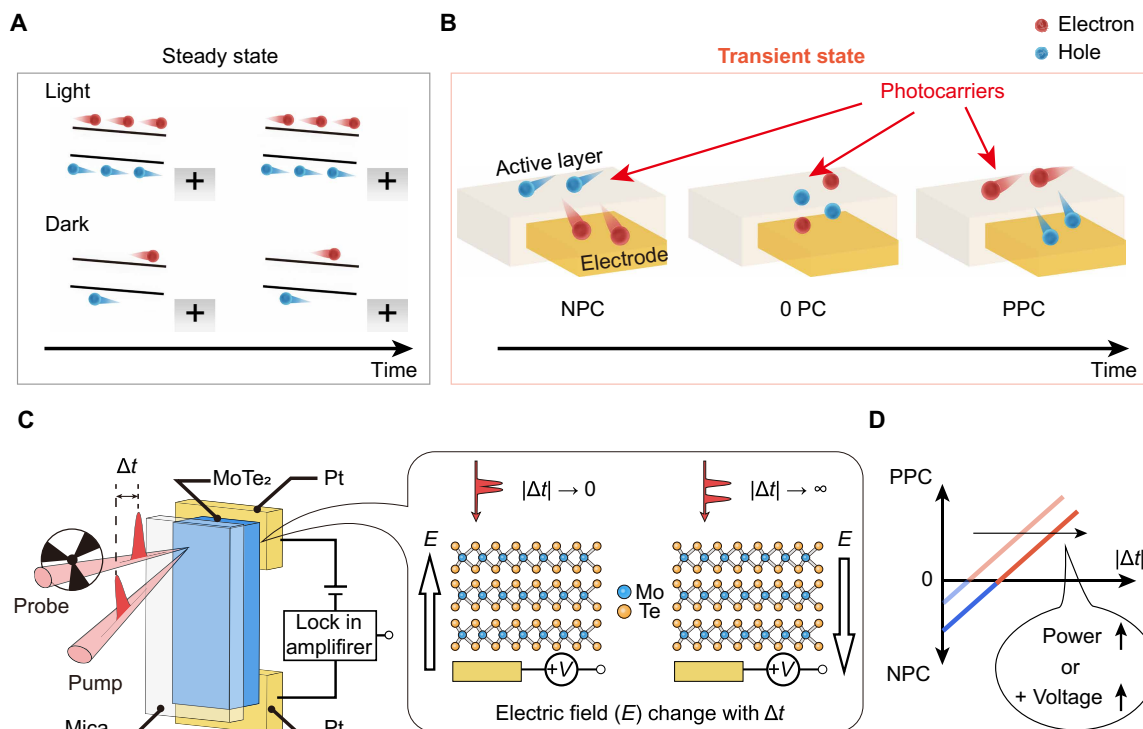
To investigate the transient electric field effects at the Schottky junction interface, we fabricated a Pt/MoTe<sub>2</sub>/Pt device using a precise

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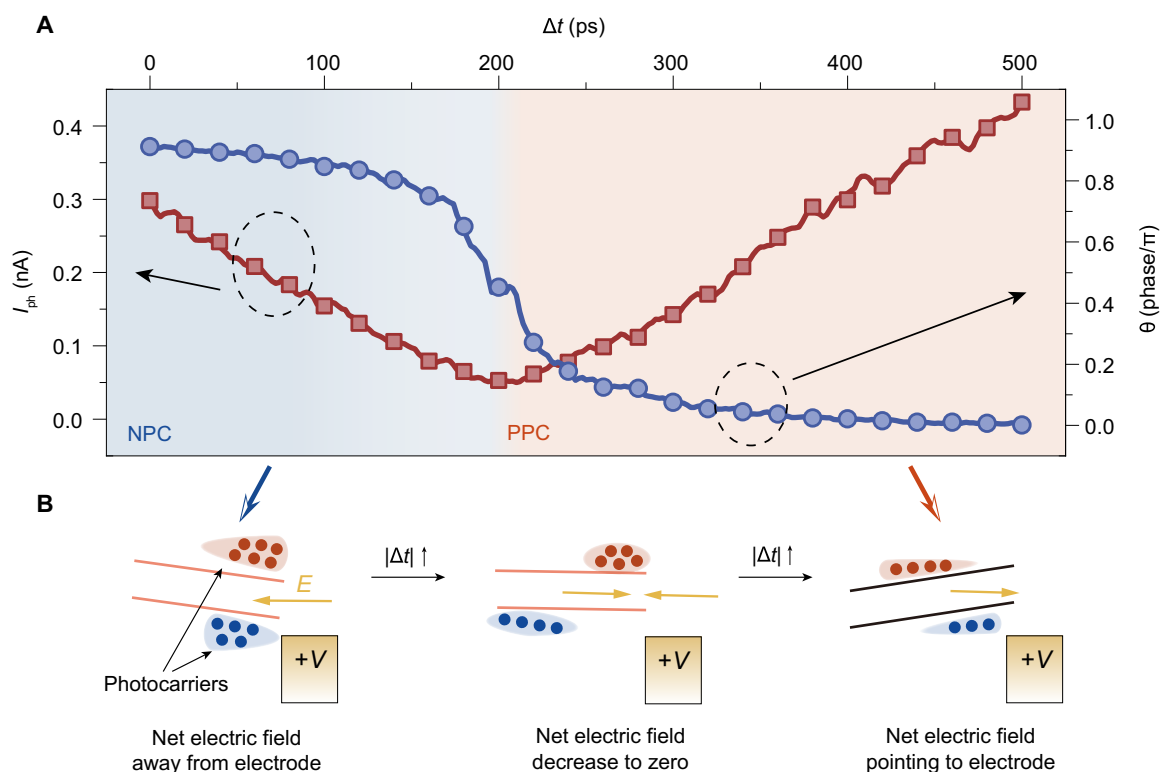
**Fig. 1. Schematic of transient electric field inversion detected from the UFPC system.** (A) Conventional time-invariant electric field in the device. (B) The transient electric field in the device. The photocarrier movement direction can be inverted with time. (C) Left: Schematic of UFPC on the device. Right: Electric field change with different  $\Delta t$  at the MoTe<sub>2</sub>/Pt interface. (D) Power and voltage manipulation for the photocurrent sign change.

tip-assisted electrode transfer method, as previously reported (Fig. 1C and fig. S1) (22). This technique preserves the van der Waals contact between the 2D material and metal electrodes (23, 24), minimizing unwanted interface states that could affect device performance. Given that platinum (Pt) has a high work function (25) compared to MoTe<sub>2</sub> (26, 27), an internal electric field forms at the Pt/MoTe<sub>2</sub> interface, oriented toward the Pt electrode. Under dark conditions, charge carriers follow the externally applied voltage, whereas, in a steady-state illuminated scenario, photocarriers predominantly migrate along the internal electric field direction when a small positive voltage is applied to Pt (28). The existence of this Schottky junction-induced electric field was confirmed via dark and light current-voltage characterization (note S2).

When a high-fluence pump light illuminates the MoTe<sub>2</sub>/Pt interface, the photocurrent polarity undergoes a remarkable inversion. This phenomenon was captured via lock-in detection, where the phase angle ( $\theta$ ) between the photocurrent signal and a fixed reference signal indicates the photocurrent direction. In our setup,  $\theta \approx 0^\circ$  corresponds to a positive photocurrent (PPC) response, whereas  $\theta \approx 180^\circ$  signifies a negative photocurrent (NPC) (Fig. 2A). Time-dependent ( $\Delta t$ ) variations in  $\theta$  confirm that photocarriers spontaneously transition from NPC to PPC. This phenomenon was also measured with a source measurement unit to exhibit more intuitive NPC/PPC inversion (fig. S13). We attribute this behavior to a light-induced inversion of the internal electric field at the Schottky interface, as illustrated in Fig. 2B. To elucidate this mechanism, we analyze the carrier dynamics at the interface. For clarity, the band alignment can be simplified to show the direction of the net electric field. Under pump light excitation, an estimated carrier density of

$3.26 \times 10^{19} \text{ cm}^{-3}$  is generated, close but higher than the intrinsic carrier density (see note S5 for detailed calculations) (29–33). At the instant of pump excitation ( $\Delta t = 0$  ps), equal numbers of photogenerated electrons and holes populate the conduction and valence bands. Initially, the net electric field direction remains unchanged, pointing toward the Pt electrode. However, as confirmed by our previous study (34), holes transfer to the metal electrode within hundreds of femtoseconds, much faster than our 2-ps measurement resolution, leaving behind an excess of electrons. This charge redistribution and light-induced Schottky barrier decrease (35) temporarily realign the electric field to reinforce the externally applied voltage direction, resulting in an initial NPC response (Fig. 2B, left). As charge recombination and carrier drift progress, a competing electric field emerges and gradually counteracts the excess electron-induced field (Fig. 2B, middle), leading to a diminishing NPC effect. Over time, as further electron recombination occurs, the internal electric field is restored to its original direction, reinstating the PPC state (Fig. 2B, right).

Alternative mechanisms, including reduced mobility, carrier density variations, or photothermal effects (36), could potentially contribute to the observed photocurrent inversion. However, systematic analysis suggests that these factors are unlikely to play a dominant role. For instance, low carrier mobility could result from a bolometric effect or exciton condensation at the interface under intense light exposure (10, 37). However, both mechanisms would accelerate carrier recombination due to reduced photocarrier diffusion lengths and mobility. If this were the dominant factor, the recombination dynamics should exhibit different time constants for small and large  $\Delta t$  values. However, our experimental data (Fig. 2A



**Fig. 2. Positive and negative photocurrent detection from UFPC.** (A) Photocurrent and corresponding phase detected from the lock-in amplifier in the UFPC system when the voltage applied to Pt is 0.11 V. (B) The principle behind the photocurrent sign change.

and fig. S15) reveal consistent recombination timescales across different  $\Delta t$  values [fitted from the exponential formula (38)], ruling out these effects. Another possibility is that density variations arise from trap states, charge redistribution to the “cold” electrode, or hot electron injection from the metal. However, several observations refute these explanations. First, given the small laser spot size ( $\sim 2\text{-}\mu\text{m}$  radius) compared to the overall device length ( $>30\text{ }\mu\text{m}$ ), local temperature uniformity is maintained, making the long-range carrier diffusion effect small (36). Thus, charge migration to distant electrodes is negligible. Second, electron injection from the Pt electrode or their departure from the interface should enhance, rather than suppress, photocurrent along the internal electric field direction. This contradicts the observed NPC response, further dismissing this hypothesis. Moreover, steady-state power-dependent photocurrent measurements (note S3) (39, 40) show no evidence of photocarrier injection from the metal electrode, reinforcing our conclusion that NPC/PPC inversion originates from transient carrier dynamics rather than extrinsic effects. Last, at small  $\Delta t$ , high-intensity pump illumination saturates trap states at the interface, reducing their influence on carrier dynamics. This further rules out a trap-mediated mechanism for NPC.

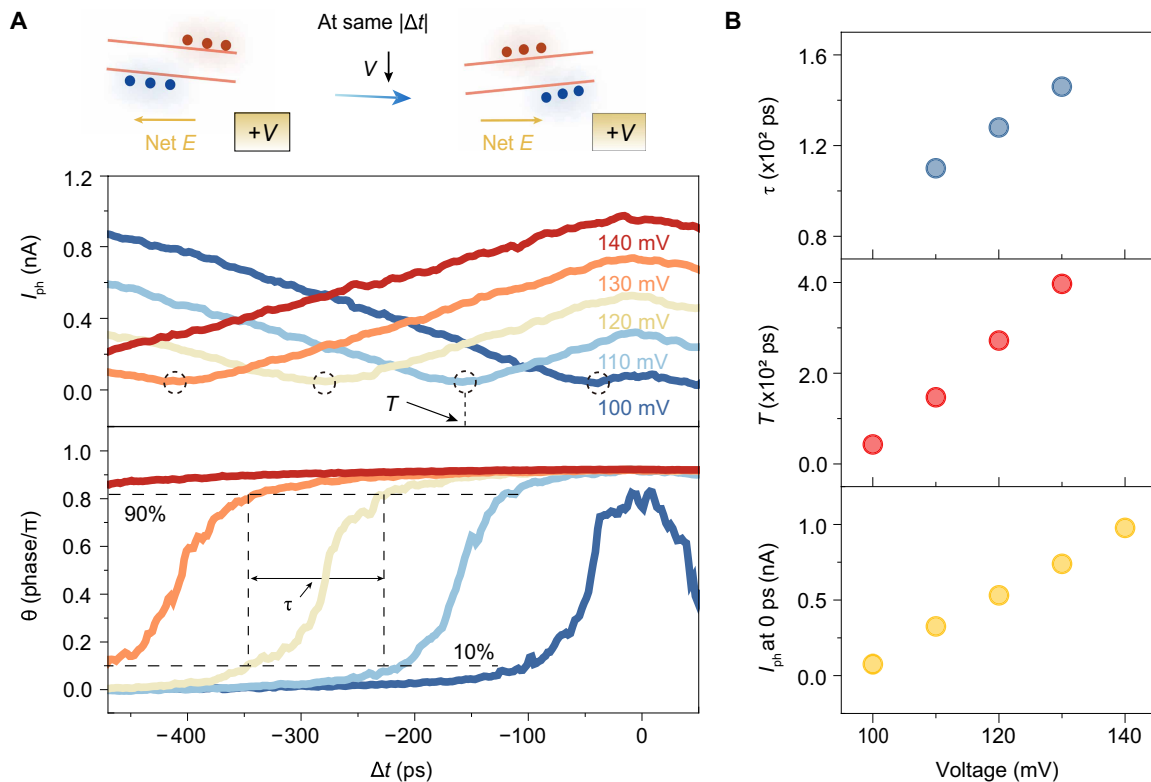
### NPC mode manipulation by the applied voltage

Given that the NPC and PPC responses stem from the interplay between the externally applied bias voltage and the internal electric field at the Schottky junction, we hypothesize that the applied voltage can modulate the NPC signal and govern the NPC/PPC inversion dynamics. To verify this, we conducted UFPC measurements under varying applied voltages (Fig. 3A). The temporal evolution of

the photocurrent ( $I_{ph}$ ) reveals a transition from NPC to PPC as  $|\Delta t|$  increases on the picosecond timescale. To quantitatively characterize this photoconductivity inversion, we define two key parameters, drawing from previous studies on ultrafast photodetectors (41, 42): (i) the duration time ( $T$ ), which corresponds to the  $\Delta t$  value at which  $I_{ph}$  reaches its minimum, and (ii) the inversion time ( $\tau$ ), defined as the  $\Delta t$  interval over which the phase angle  $\theta$  transitions between 90 and 10% of its total variation (maximum  $\theta$  – minimum  $\theta$ ) (Fig. 3A, bottom). These parameters provide a straightforward comparison of the speed of the inversion process.

As the applied voltage increases, we observe a simultaneous increase in  $T$ ,  $\tau$ , and the  $I_{ph}$  at  $\Delta t = 0$  ps (Fig. 3B). This trend can be attributed to the reduction in the Schottky barrier height under higher applied voltages (Fig. 3A, top). As discussed previously, the redistribution and recombination of photocarriers at the interface gradually weaken the net electric field pointing toward the 2D material, until equilibrium is reached. Once the density of photocarriers declines sufficiently, the net electric field direction reverts to its original orientation, aligning with the internal electric field of the Schottky junction. Throughout this process, higher applied voltages decelerate the transition from NPC to the equilibrium state by reducing the driving force along the internal electric field and favoring carrier movement in the direction of the applied voltage. Consequently, larger applied voltages lead to increased  $\tau$  values, indicating a slower NPC-to-PPC transition. Furthermore, under stronger external bias, fewer photocarriers are required to induce electric field inversion, thereby prolonging  $T$  as well.

Our explanation is further corroborated by power-dependent UFPC measurements and Schottky barrier engineering experiments.



**Fig. 3. Voltage manipulation of the NPC and PPC.** (A) Top: The principle of voltage manipulation. Middle: UFPC results from different applied voltages. Bottom: Phase change. (B) Transfer time (top) from NPC to PPC, duration time (middle) of NPC, and photocurrent (bottom) when  $\Delta t = 0$  ps.

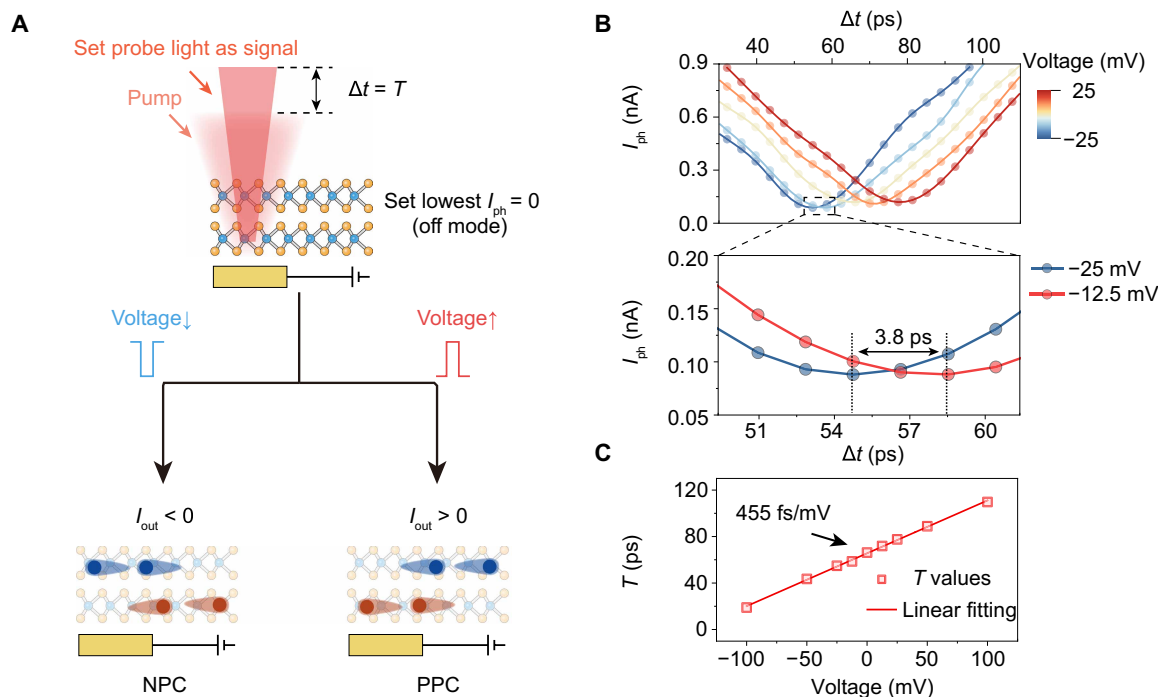
Given that higher excitation power generates a greater carrier density, it effectively reduces the net electric field along the internal electric field direction, mimicking the effects of increased applied voltage. As shown in fig. S18, high-power excitation results in larger  $T$  and  $\tau$  values compared to lower-power excitation, reinforcing our hypothesis. Beyond power-dependent measurements, we also investigated the influence of Schottky barrier height on the NPC/PPC inversion using different metal contacts on the same MoTe<sub>2</sub>-based device. By replacing Pt with silver (Ag), a lower work function metal, we systematically modulated the Schottky barrier at the interface (fig. S19). Given Ag's smaller work function (43), the Ag/MoTe<sub>2</sub> device exhibited a slower inversion speed ( $1/\tau$ ) and reduced inversion voltage compared to the Pt/MoTe<sub>2</sub> device. This suggests that a lower Schottky barrier results in a more gradual NPC-to-PPC transition. These findings further validate that the NPC/PPC inversion originates from net electric field modulation at the metal/2D material interface.

### Precisely voltage-tuned NPC/PPC photodetector

PICs require photodetectors capable of operating in both NPC and PPC modes to perform complex logic operations (44, 45). For high-speed optoelectronic applications, two crucial aspects must be considered: (i) rapid transition from an off-state to an active working mode and (ii) ultrafast inversion between NPC and PPC states. Given the tunability of the duration time ( $T$ ) and the nearly constant minimum  $I_{ph}$  across different applied voltages, our ultrafast photodetector design can satisfy both criteria. For the first aspect, to achieve rapid state activation, we propose a precisely voltage-tuned NPC/PPC photodetector that leverages the controllable delay

time ( $\Delta T$ ) of the photocurrent signal. As illustrated in Fig. 4A, when a probe light pulse arrives at the device after a fixed delay ( $\Delta t$ ), the corresponding  $I_{ph}$  response is modulated by the applied voltage. At  $\Delta t = T$ ,  $I_{ph}$  reaches its minimum, which can be designated as the off-state of the photodetector, effectively setting the output photocurrent ( $I_{out}$ ) to zero. By increasing (or decreasing) the applied voltage,  $I_{out}$  acquires a negative (or positive) value relative to this offset, enabling selective operation in NPC or PPC modes without additional structural modifications (Fig. 4A). Direct measurement of a subpicosecond voltage-to-photocurrent response remains a challenge due to current technological limitations. However, the similar shape of UFPC curves across different applied voltages allows us to use the shift in  $T$  (denoted as  $\Delta T$ ) to estimate the transition speed from off-mode to active operation. As shown in Fig. 4B, our best-performing device exhibits a  $\Delta T$  of 3.8 ps, corresponding to a picosecond-level ( $\sim 250$  GHz) programmable logic operation (Fig. 4B, bottom). Linear fitting of  $T$  versus  $V$  (Fig. 4C) yields a slope of 454 fs/mV, suggesting that, in an ideal case, a 1-mV voltage stimulus could induce a 454-fs ( $>1$ -THz) response. While achieving this theoretical limit remains challenging due to surface states, interface defects, and variations in metal-semiconductor contact quality, advancements in device fabrication and the integration of short-lifetime materials could push this response time into the femtosecond regime.

Beyond fast activation, the photodetector must also switch rapidly between NPC and PPC states. Figure 4B shows that the  $I_{ph}$  curve for NPC under higher applied voltage intersects with the  $I_{ph}$  curve for PPC under lower applied voltage at a specific  $\Delta t$ . This



**Fig. 4. Ultrafast device based on electric field-induced photocurrent sign change.** (A) Working principle of the photodetector. (B) Top: UFPC results from the fastest response device. Bottom: Enlarged figure of  $-25$  mV and  $-12.5$  mV UFPC results for the lowest  $I_{ph}$ . The lowest  $I_{ph}$  time difference is 3.8 ps. (C) Linear fitting for the lowest  $I_{ph}$  from different voltages.

suggests that NPC/PPC inversion is governed solely by the electric field reconstruction at the metal/2D material interface, which occurs on a subpicosecond timescale ( $<1$  ps) (15, 16). Unlike conventional methods relying on gate voltage modulation (44) or pump-induced carrier dynamics (21), this approach enables ultrafast photoconductivity inversion without requiring additional external tuning elements. The switching speed is inherently limited only by the intrinsic speed of electric field reconfiguration or optical intensity modulation, making it significantly faster than previous techniques.

Achieving ultrafast, precisely voltage-controlled NPC/PPC switching holds immense potential for next-generation optoelectronic devices. Such a photodetector architecture could be integrated into ultrafast optical switches (46), photoelectric logic devices (47), or artificial synapses (48), offering unprecedented speed and energy efficiency. For example, NPC/PPC inversion driven by ferroelectric polarization requires prior carrier accumulation and subsequent polarization switching prior to the reversal of current polarity (44). These prerequisite steps inevitably extend the overall response time necessary to achieve inversion, even though light-induced polarization has been reported to occur within hundreds of femtoseconds (49) and device-level switching responses typically operate on the microsecond scale (50). In contrast, our device achieves polarity inversion without these intermediate steps, allowing for a complete response within the picosecond regime, with the potential to reach femtosecond timescales in future implementations. By leveraging the intrinsic electric field dynamics of Schottky-contacted 2D materials, this design circumvents the limitations of traditional photodetectors, paving the way for high-performance, reconfigurable photonic circuits.

## DISCUSSION

In conclusion, we have demonstrated the dynamic manipulation of a photoinduced transient electric field at the  $\text{MoTe}_2/\text{Pt}$  interface in a  $\text{Pt}/\text{MoTe}_2/\text{Pt}$  Schottky junction device, unveiling a voltage-dependent ultrafast photoconductivity transition from NPC to PPC. By precisely controlling the applied electric field on a picosecond timescale, we achieved a seamless inversion of photoconductivity as a function of  $\Delta t$ , with tunable polarity switching governed by external voltage modulation. Leveraging this mechanism, we proposed and realized a voltage-programmable NPC/PPC photodetector, achieving an exceptional 3.8-ps response time under a minimal voltage stimulus of just 12.5 mV. This breakthrough eliminates the need for additional modulation components, dramatically enhancing both response speed and operational simplicity. The demonstrated capability of ultrafast, electrically tunable photoconductivity switching paves the way for next-generation PICs, high-speed optical communication systems, ultrafast imaging technologies, and quantum information processing, offering new avenues for the development of high-performance optoelectronic and photonic devices.

## MATERIALS AND METHODS

### Device fabrication

The bulk single crystal of  $\text{MoTe}_2$  (2D semiconductor) is used to obtain 50-nm sample of  $\text{MoTe}_2$  with mechanical exfoliation. The  $\text{MoTe}_2$  was cleaved on the tape and directly attached to the mica substrate (Changchun City Taiyuan Fluorophlogopite Co. Ltd.). The 50-nm sample was selected with optical microscopy and atomic force microscopy (Hitachi, AFM5000II). After that, Pt electrodes were transferred using the tip-assistant method (note S1). For the

metal electrode change on the same sample, a similar procedure was applied to the same MoTe<sub>2</sub> sample twice.

### Ultrafast photocurrent measurement

The ultrafast photocurrent measurement was performed on a home-built setup (fig. S1). An ~13-fs pulse laser (Coherent Vitara-T, 80-MHz repetition rate) centered at 800 nm was split into two beams by a 50/50 beam splitter to form the pump-probe configuration. To avoid an interference effect, the polarization direction of the pump beam was rotated to be perpendicular to that of the probe beam with a half-waveplate. The delay time of the probe beam was achieved from different path lengths, precisely controlled by a mechanical delay stage (Newport, UTS150CC and ESP301 controller). This light beam was mechanically chopped at 100-Hz frequency. The pump and probe beams were recombined by another beam splitter and focused on the device through a microscope objective (Mitutoyo, MY50X-825) to generate photocurrent, which was measured with a preamplifier (Keithley, 2634B) and a lock-in amplifier (Stanford, SR830) synchronized with the mechanical chopper.

### Supplementary Materials

This PDF file includes:

Notes S1 to S5

Figs. S1 to S21

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